Capstone Project Document

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Project Title :
Product Sales Data Analysis using Power BI

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1. Project Overview

Businesses must maintain detailed records of their sales and financial performance. Analyzing this data helps organizations understand their financial health, identify profitable products and regions, and make strategic decisions.
This capstone project focuses on analyzing Microsoft's Financial Sample Dataset using Power BI to evaluate sales performance, profit distribution, and business trends. The project involves cleaning raw data, creating meaningful visualizations, and developing an interactive dashboard that supports business decision-making.

Dataset to be Used - You can check in the Resource Section

Financial Sample.xlsx

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2. Project Objectives

The key objectives of this project are:

- To analyze sales and profit performance across different products and regions
- To identify trends and patterns in profitability over time
- To determine top-performing and low-performing products
- To evaluate regional contribution to overall business revenue
- To create an interactive Power BI dashboard
- To gain hands-on experience in real-world data analytics

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3. Dataset Details

Dataset Name: Microsoft Financial Sample Dataset
File Type: Excel Workbook
Source: Microsoft Power BI Sample Data

Key Fields in Dataset:

- Segment
- Country
- Product
- Discount Band
- Units Sold
- Manufacturing Price
- Sale Price
- Gross Sales
- Discounts
- Sales
- COGS
- Profit
- Date

This dataset contains structured financial data that enables detailed analysis of sales and profit by country, product, and segment.

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4. Tools and Technologies Used

- Microsoft Power BI Desktop
- Power Query Editor
- Microsoft Excel
- DAX (Data Analysis Expressions)
- Data Visualization Techniques

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5. Project Workflow

The project is divided into multiple small tasks to ensure systematic execution.

Phase 1 – Project Setup

Task 1: Understanding Business Requirement

- Understand project goals
- Identify business problems
- Define KPIs to be analyzed

Task 2: Dataset Acquisition

- Download the Financial Sample Excel dataset
- Review dataset structure
- Understand available fields and data types

Phase 2 – Data Import and Preparation

Task 3: Import Data into Power BI

- Open Power BI Desktop
- Use “Get Data → Excel” option
- Load the Financial Sample workbook

Task 4: Data Exploration

- Review all available columns
- Identify missing or incorrect values
- Understand the structure of data

Task 5: Data Cleaning using Power Query

- Rename unclear columns
- Remove duplicate records
- Handle blank or null values
- Correct data types
- Format date column

Task 6: Data Transformation

- Create additional calculated columns if needed
- Extract Year and Month from Date
- Standardize product and segment names
- Categorize profit ranges

Phase 3 – DAX Measures Creation

Task 7: Create DAX Measures

Create important calculated measures such as:

- Total Sales
- Total Profit
- Total Units Sold
- Average Discount
- Profit Margin
- Year-to-Date Sales
- Year-to-Date Profit
- Previous Year Profit
- Growth Percentage

Phase 4 – Dashboard Development

Task 8: Profit Trend Analysis

- Create Line Charts to track:
  - `Monthly profit trends`
  - `Yearly profit trends`
- Analyze seasonality and growth patterns

Task 9: Regional Profit Analysis

- Build Bar Charts to show:
  - `Profit by Country`
  - `Profit by Segment`
- Compare performance across regions

Task 10: Global Profit Distribution

- Add a World Map Visual
- Display profit distribution geographically

Task 11: Product Performance Analysis

- Use Clustered Column Charts
- Highlight:
  - `Top performing products`
  - `Bottom performing products`

Task 12: Category-Wise Performance

- Use Treemap or Stacked Bar Visuals
- Analyze strong and weak product segments

Task 13: KPI Visualization

Create KPI cards for:

- Total Sales
- Total Profit
- Total Units Sold
- Profit Margin
- Average Discount

Task 14: Filters and Interactivity

Add slicers for:

- Country
- Year
- Product
- Segment

Phase 5 – Dashboard Finalization

Task 15: Formatting and Design

- Apply consistent theme
- Format visuals properly
- Add titles and labels
- Improve layout and alignment

Task 16: Testing Dashboard

- Validate all calculations
- Cross-check values
- Test slicers and filters

Task 17: Final Report Preparation

- Export dashboard
- Document insights
- Prepare final project summary

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6. Expected Outcomes

After completing this project, the following outputs will be delivered:

- Fully interactive Power BI dashboard
- Clean and structured dataset
- Visual insights on sales and profitability
- Business performance analysis
- Actionable recommendations

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7. Business Benefits

This analysis helps businesses to:

- Identify profitable products and regions
- Track financial performance over time
- Make informed strategic decisions
- Improve sales and marketing strategy
- Optimize discount and pricing strategies

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8. Analytical Questions

The dashboard will help answer the following key business questions:

Sales Performance

1. Which country generates the highest profit?
2. Which product has the highest sales?
3. Which segment contributes most to revenue?
4. What is the overall profit margin?
5. Which month shows the highest sales?

Trend Analysis

6. How has profit changed year over year?
7. Are there seasonal patterns in sales?
8. Which year performed best?

Product-Level Insights

9. Which products are top performers?
10. Which products are least profitable?
11. Which discount band generates maximum sales?

Regional Insights

12. Which regions need improvement?
13. What is the profit contribution by country?
14. Which segment performs best in each region?

Strategic Business Questions

15. Where should the company focus more investment?
16. Which products should be promoted?
17. Which products may need discontinuation?
18. Does higher discount increase profitability?
19. Which product-country combination is most profitable?
20. What are the key drivers of profit?

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9. Conclusion

This capstone project demonstrates how business data can be transformed into meaningful insights using Power BI. By analyzing sales and profit trends, the organization can identify strengths, weaknesses, and growth opportunities.

The project provides practical exposure to:

- Data cleaning
- Data transformation
- Data visualization
- Business analytics
- Dashboard creation

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10. Future Enhancements

This project can be further enhanced by:

- Adding sales forecasting
- Connecting real-time data sources
- Implementing advanced DAX analytics
- Integrating customer behavior analysis
- Building automated refresh dashboards

SUBMISSION REQUIREMENTS

Create one main folder named:
Capstone_Project_YourFullName
Example: Capstone_Project_RohitRaj

Inside this folder, include the following files:

1. Power BI (.pbix) File
2. Documentation File (Word or PDF)
   The documentation must include:
   - Screenshots of Data Import
   - Screenshots of Power Query Cleaning
   - Screenshots of DAX Measures
   - Screenshots of Dashboard Pages
   - Answers to all Analytical Questions
   - Proper Part Numbers and Question Numbers

FINAL SUBMISSION FORMAT

- Create one folder
- Put all required files inside
- Convert folder to ZIP
- Upload only ZIP

ZIP must contain (as applicable):

- Screenshots
- Image (if used)

- Doc file
- PDF file
- Code file / MD file (if used)
- Excel file (if used)
- Power BI (.pbix) file

Do NOT:

3. Upload individual files
4. Upload multiple ZIP files

Watch this to learn how to create ZIP file

https://youtu.be/c7x-Q3AQEgw?si=_mS4c7OXADQ7DroE

If you have any doubt, you can directly contact your mentor.